2025 IEEE Workshop on Applications of Signal Processing to Audio and Acoustics (WASPAA) | 979-8-3315-3745-6/25/$31.00 ©2025 IEEE | DOI: 10.1109/WASPAA66052.2025.11230965

OpenBEATs: A Fully Open-Source General-Purpose Audio Encoder

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***Abstract*—Masked token prediction has emerged as a powerful pre-training objective across language, vision, and speech, offering the potential to unify these diverse modalities through a single pre-training task. However, its application for general audio understanding remains underexplored, with BEATs being the only notable example. BEATs has seen limited modifications due to the absence of open-source pre-training code. Furthermore, BEATs was trained only on AudioSet, restricting its broader downstream applicability. To address these gaps, we present OpenBEATs, an open-source framework that extends BEATs via multi-domain audio pre-training. We conduct comprehensive evaluations across six types of tasks, twenty five datasets, and three audio domains, including audio reasoning tasks such as audio question answering, entailment, and captioning. OpenBEATs achieves state-of-the-art performance on six bioacoustics datasets, two environmental sound datasets and five reasoning datasets, performing better than models exceeding a billion parameters at one-fourth their parameter size. These results demonstrate the effectiveness of multi-domain datasets and masked token prediction task to learn general-purpose audio representations. To promote further research and reproducibility, we release all pre-training and evaluation code, pretrained and fine-tuned checkpoints, and training logs [1].**

1. INTRODUCTION

Self-supervised learning (SSL) has shown significant promise across a wide range of audio processing tasks. It allows models to learn general-purpose representations that transfer effectively to various downstream applications. Notable examples of SSL-based audio encoders (AEs) include BEATs [1], SS-AST [2], MAE-AST [3], and Audio-MAE [4]. Among these, BEATs stands out due to the wide adoption of its pre-trained checkpoints, strong performance across DCASE challenges [5]–[7] and the potential to unify vision, language and audio encoders through masked token prediction based learning. However, the pre-training pipeline for BEATs remains closed-source, limiting such broader research impact. In addition, BEATs uses a multistage pre-training setup, combining teacher-student distillation and masked audio modeling, making the pre-training more complex than other AEs. In the speech community, open-sourcing models and their training pipelines has been a crucial practice [8]–[11] that leads to enhanced reproducibility and accessibility. Motivated by these gaps, this work aims to completely open-source the BEATs pre-training using the ESPnet toolkit [12], [13] to foster further advancements.

Despite advances in SSL-based audio modeling, the training and evaluation of audio encoders remain fragmented across distinct domains – environmental sound, bioacoustics and music. As a result, state-of-the-art (SOTA) performance is typically achieved by domain-specific models. For instance, BEATs excels in diverse environmental sound benchmarks, GPM-BT [14] leads in bioacoustics tasks, and MERT [15] sets the bar for music-related tasks. In contrast, the speech community has demonstrated that multitask and multilingual training [16]–[18] produces more generalizable and transferable representations. Similar trends are evident in Audio-Language Models (ALMs), which unify diverse audio tasks under a shared sequence-to-sequence framework. The performance of ALMs [19]–[21] is often strongly correlated with the quality of the underlying audio encoder [22], underscoring the need for robust, general-purpose encoders. Yet, the current literature lacks both a universal audio encoder and a standardized cross-domain benchmark for evaluating generalization. To address these limitations, we present a unified encoder trained on diverse audio domains. Our results verify that multi-domain pre-training improves cross-domain generalization, achieving SOTA performance on multiple datasets.

Evaluation of audio encoders has largely been limited to classification tasks on a narrow set of general-purpose datasets, such as AudioSet [23] and ESC-50 [24]. Broader evaluations – particularly in specialized domains like bioacoustics – remain underexplored or are deferred to downstream users. However, with the advent of ALMs, audio encoders are increasingly being used for more complex semantic tasks, such as audio captioning, audio question answering, and audio reasoning, spanning diverse domains. In this setup, the audio encoder is expected to contain both audio information and necessary semantic information to steer the language model. This shift highlights the need for comprehensive evaluation of audio representations not only on traditional classification tasks, but also on open-ended and semantically rich tasks across bioacoustics, environmental sound, and music domains. To support this, we benchmark multiple models and offer a streamlined interface, via a single script, for testing new encoders using ESPnet.

In this work, we address the aforementioned challenges, by releasing an open-source and reproducible training pipeline for BEATs, scaling it across diverse domains, and proposing a unified benchmark for comprehensive ALM-focused evaluation. The main contributions, also depicted in Fig. 1, are:

- We open-source the training and evaluation framework for BEATs, a SOTA SSL audio encoder.
- We scale the BEATs model to achieve general-purpose audio representations that can be used for a variety of downstream tasks. Specifically, we increase the model size to 300M parameters and expand the pre-training data to 20k hours by adding environmental sound datasets and incorporating bioacoustics and music domains. Our model achieves strong cross-domain transfer and is comparable to models with over a billion parameters [25].
- We introduce an open-source comprehensive evaluation suite that spans multiple domains, including bioacoustics, music and environmental sound. Our benchmark is designed to streamline the evaluation of audio encoders and include tasks such as audio entailment, question answering, and captioning.

2. BACKGROUND

2.1. BEATs Architecture

BEATs (Bidirectional Encoder representation from Audio Transformers) [1] is based on a Vision Transformer [26] architecture adapted for audio representation learning. The model processes 16 kHz audio

[1]https://shikhar-s.github.io/OpenBEATs

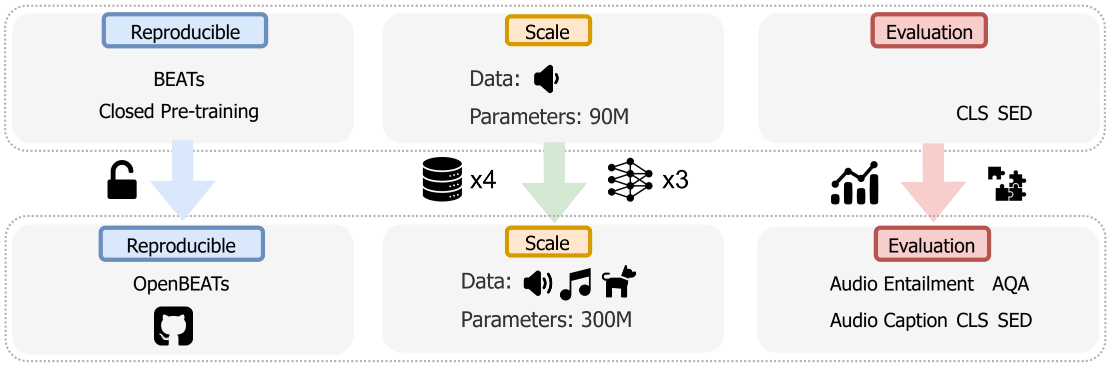

Fig. 1: Our contributions in this work are three folds: 1) Open sourcing the pre-training and evaluation code for BEATs. 2) Scaling the training data and model to handle multiple sound domains like music (), environmental sound () and bioacoustics (). 3) Fine-grained multi-domain evaluations that go beyond standard tasks, probing reasoning ability via audio entailment and question-answering.

by converting it into 128-dimensional mel spectrograms, which are segmented into 16×16 patches and then fed to a transformer encoder. This patching mechanism enables BEATs to capture both local and global spectral features efficiently. The backbone of the model consists of stacked transformer encoder blocks, each employing multi-head self-attention to learn contextual representations of audio signals.

2.2. BEATs Pre-training

BEATs undergoes two-stage SSL, involving encoder training and tokenizer training. These stages iteratively refine both the feature representations and the discrete tokenization of audio signals, improving the model's ability to learn structured embeddings, with the process repeated up to three times.

2.2.1. Encoder Training: The pre-training of BEATs encoder follows a Masked Language Modeling (MLM) paradigm applied to audio. In the first iteration, masked spectrogram patches are tokenized using a randomly initialized tokenizer [27]. From the second iteration onward, the model employs a trained tokenizer, leading to progressively refined learned representations. Given an input sequence of d dimensional Mel-spectrogram patches with length N, $\mathbf{X} = \{\mathbf{x}_n \in \mathbb{R}^d\}_{n=1}^N$, a subset of patches is randomly masked, producing a sequence of unmasked patches $\tilde{\mathbf{X}}$. The encoder produces a sequence of h-dimensional embeddings $f^e(\tilde{\mathbf{X}}) = \{\mathbf{o}_n \in \mathbb{R}^h\}_{n=1}^N$, and it is fed through the label predictor module f^e_{pred}. Then, $f^e_{\text{pred}} \circ f^e$ is learned to predict the masked tokens by minimizing the cross-entropy loss between the predicted tokens and the ground truth token indices given the unmasked patches:

$$\mathcal{L}_{\text{MLM}} = -\sum_{n \in M} \log p(z_n|\tilde{X}), \tag{1}$$

where M is the set of masked positions and z_n represents the ground truth token for position n yielded from the tokenizer. Therefore, $\log p(z_n|\tilde{X})$ stands for the predicted probability distribution of p over possible tokens at position n, given the masked input $\tilde{\mathbf{X}}$. At the implementation level, besides tracking the loss in (1), metrics like vocabulary coverage and masked accuracy are also tracked in our open-source implementation.

2.2.2. Tokenizer Training: BEATs adopts a teacher-student framework for tokenizer training. The teacher model corresponds to the encoder above from the previous iteration, while the student model is the tokenizer being trained in the current iteration.

The tokenizer encodes input $\mathbf{X}$ into embedding sequence $f^t(\mathbf{X}) = \mathbf{E} = \{\mathbf{e}_n \in \mathbb{R}^p\}_{n=1}^N$, which is quantized into $q(\mathbf{E}) = \{\mathbf{v}_{z_n} \in \mathbb{R}^p\}_{n=1}^N$. Here, the token z_n is the nearest neighbor index in the tokenizer codebook $\mathbf{V} = \{\mathbf{v}_1, \mathbf{v}_2, \cdots, \mathbf{v}_K\}$ of size K:

$$z_n = \arg\min_t |\bar{\mathbf{v}}_t - \bar{\mathbf{e}}_n|_2^2\,, \tag{2}$$

where $\bar{\mathbf{v}}_t$ and $\bar{\mathbf{e}}_n$ is the L2-normalized $\mathbf{v}_t$ and $\mathbf{e}_n$.

Given the quantized embedding sequence $q(\mathbf{E})$, the tokenizer learns to predict embeddings $f^t_{\text{pred}}(q(\mathbf{E})) = \{\hat{\mathbf{o}}_n \in \mathbb{R}^p\}_{n=1}^N$ to align with the teacher embeddings through the knowledge distillation loss $\mathcal{L}_{\text{KD}}$:

$$\mathcal{L}_{\text{KD}} = -\sum_n \cos(\hat{\mathbf{o}}_n, \mathbf{o}_n) + |\text{sg}[\bar{\mathbf{e}}_n] - \bar{\mathbf{v}}_{z_n}|_2^2 + |\bar{\mathbf{e}}_n - \text{sg}[\bar{\mathbf{v}}_{z_n}]|_2^2\,, \tag{3}$$

where $\text{sg}[\cdot]$ is the stop gradient operator. Equation (3) has three terms: a cosine similarity loss encouraging alignment between the teacher and tokenizer, and two L2 loss terms improving stability and consistency of codebook representations. At the implementation level, the gradients of the quantized embedding sequence $q(\mathbf{E})$ are copied to the embedding sequence $\mathbf{E}$ to solve the non-differentiable nature of vector quantization [28], and the codebook $\mathbf{V}$ is initialized using K-Means and optimized through exponential moving average.

3. OpenBEATs

In this work, we extend the pre-training data beyond AudioSet [23] by incorporating additional large-scale public datasets such as FreeSound, FMA [29], and iNaturalist [30], which span diverse domains including music, environmental sound, and bioacoustics. In prior literature [14], [31] bioacoustics datasets have been used in isolation and not combined with music. However, to the best our knowledge, this work is the first to systematically unify these diverse domains and study the cross-domain impact from such multi-domain training via extensive evaluations. In addition, our model is trained with variable-length input sequences, making it more adaptable to the natural variability of real-wold audio data. Together, these changes enable our encoder to learn robust, general purpose representations for a wide range of tasks and domains. A detailed list of our pre-training datasets is provided in Table 2. We exclude speech corpora, as speech requires fine-grained temporal resolution to model phoneme durations, which differs from the coarser time resolution of 175 ms employed in a BEATs patch.

We first train a 90M parameter OpenBEATs model (`base`) on this expanded dataset and observe saturation in downstream task performance. To address this, we scale up the model architecture and train a 300M parameter variant (`large`), following the `base`/`large` design pattern established by HuBERT [11]. Table 1 lists the details of our architecture and pre-training hyper-parameters. To support

Table 1: Architecture, data, and training configuration for BEATs and OpenBEATs variants.

Model	Data (hr)	Param	Hidden size	FFN size	Layers	# Heads	# Codebook	Batch (sec)	Updates	Warmup	LR	Opt.
BEATs	5.8k	90M	768	3072	12	12	1024	4.6k	400k	32k	5e-4	AdamW
OpenBEATs -Base	20k	90M	768	3072	12	12	1024	10.7k	400k	40k	5e-4	AdamW
OpenBEATs -Large	20k	300M	1024	4096	24	16	1024	10.7k	400k	40k	1e-4	AdamW

reproducibility and future research, we open-source the full training pipeline, including the pre-training code, data preprocessing scripts, and training logs for all our models.

Table 2: Overview of pre-training datasets covering music (♫), bioacoustics (🐕) and environmental sound events (🔈). *We use data filtered using the WavCaps [32] processing pipeline for these sources.

Dataset	Hours	Instances	Domain
FMA [29]	7.7k	2.8M	♫
AudioSet [23]	5.6k	2.0M	🔈
FreeSound*	4.1k	1.7M	🔈
BBC Sound Effects*	1.0k	0.4M	🔈
iNat Sounds [30]	0.8k	0.3M	🐕
Others [33], [34]	0.3k	0.1M	🔈
Total	**20k**	**7.3M**	

Table 3: Evaluation tasks. Recipes for all of these tasks are also open-sourced as part of this work. LP = Linear Probing; FT = Full Fine-tuning; DFT = Decoder Fine-Tuning; CLS = Classification; SED = Sound Event Detection. We follow prior works for evaluation protocols where available (eg. BEANS). Higher values are better for all metrics. See details in Section 4.2.

Dataset	Abbr.	Task Type	Eval	Metric	Reference
Environmental Sound					
DESED	DSD	SED	LP	Score	[35]
UrbanSound8K	US8K	CLS	LP	Score	[36]
FSD-50K	F50K	SED	LP/FT	Score/mAP	[37]
ESC-50	ESC	CLS	LP/FT	Score/Acc	[24]
FSD2018-Kaggle	FKgl	SED	LP	Score	[38]
Clotho (Retrieval)	Clotho	Retrieval	LP	Score	[39]
AudioSet-2M	AS-2M	SED	FT	mAP	[23]
AudioSet-20K	AS-20K	SED	FT	mAP	[23]
Bioacoustics					
BEANS (10 tasks)	BEANS	CLS/SED	FT	Acc/mAP	[40]
Reasoning					
Audio Entailment	Entailment	Entailment	FT	Acc	[41]
AudioQA	AQA	Question Answering	FT	Acc	[42]
Clotho (Captioning)	Clotho AAC	Captioning	DFT	CIDEr [43]	[39]
Music					
GTZAN	GTZAN	CLS	FT	Acc	[44], [45]
NSynth-Instr.	NSynth-I	CLS	FT	Acc	[46]
NSynth-Pitch	NSynth-P	CLS	FT	Acc	[46]

4. EVALUATION SETUP

4.1. Baselines

Dasheng: Dasheng [25] surpasses the billion-parameter mark and is pre-trained on over 272k hours of data, covering multi-domain and speech-rich sources from YouTube. In contrast, our setup offers a more controlled comparison by deliberately restricting to audio-only datasets. Dasheng adopts a mean squared error loss, whereas OpenBEATs relies on masked modeling with discrete tokens.

Speech models: To gauge the transferability of speech-only pre-training, we compare with Data2Vec [48] and Whisper-small [49].

Bioacoustics models: We evaluate domain-specific pre-training via dedicated bioacoustics sound encoders like AVES [52], BioLingual [31], to test the claim that multi-domain SSL can outperform models trained on specific datasets. We also report GPM-BT [14] which has shown promising results on bioacoustics. While GPM-BT and AVES are SSL models, BioLingual is trained with text supervision following a CLAP-like training methodology.

Table 4: Linear probing results on sound tasks from X-ARES benchmark [47]. `B` and `L` denotes `Base` and `Large` models. OpenBEATs achieves best performance on DESED and UrbanSound-8K datasets, and performs competitively with other models of similar scale across the remaining datasets.

Model	Param	DSD	US8k	F50k	ESC	FKgl	Clotho	Avg
Dasheng `B` [25]	90M	0.53	0.84	0.41	0.87	0.56	0.03	0.54
Data2Vec [48]	90M	0.14	0.44	0.08	0.25	0.20	0.01	0.19
Whisper [49]	74M	0.13	0.72	0.26	0.61	0.48	0.03	0.37
EAT `B` [50]	88M	0.36	0.79	0.36	0.66	0.38	0.04	0.43
BEATs (iter3) [1]	90M	0.56	0.85	0.22	0.84	0.55	0.04	0.51
EAT `L` [50]	309M	0.44	0.82	0.44	0.73	0.58	**0.06**	0.51
Dasheng-0.6B [25]	600M	0.54	0.85	0.45	0.88	0.58	0.04	0.56
Dasheng-1.2B [25]	1.2B	0.56	0.85	**0.46**	**0.89**	**0.63**	0.04	0.57
OpenBEATs `L`	300M	**0.57**	**0.87**	0.43	0.86	0.54	0.05	0.55

Table 5: OpenBEATs achieves best performance on environmental sound detection tasks. † denotes values reported by BEATs, all other values are from our open-source implementation. Please note that our copy of AudioSet (AS) downloaded from YouTube differs from BEATs. See Section 5 for details.

Model	Param	ESC-50 acc	FSD-50K mAP	AS-2M mAP	AS-20K mAP
Audio-MAE [4]	86M	94.1†	-	47.3†	37.1†
Audio-MAE	304M	-	-	47.4†	37.6†
BEATs iter 1	90M	93.0†	52.3	47.9†	36.0†
BEATs iter 2	90M	95.1†	55.4	48.1†	36.0†
BEATs iter 3	90M	95.6† / 94.8	56.2	48.0† / 41.6	38.3† / 34.1
OpenBEATs iter 1	90M	93.9	54.4	39.4	31.5
OpenBEATs iter 2	90M	94.8	55.8	41.1	31.5
OpenBEATs iter 3	90M	95.0	55.5	40.8	32.1
OpenBEATs iter 1	300M	95.3	56.7	41.5	32.7
OpenBEATs iter 2	300M	95.7	**57.5**	42.2	32.4
OpenBEATs iter 3	300M	**95.8**	56.8	42.1	33.1

AudioMAE: AudioMAE employs a pure mask-and-reconstruct objective. This is an alternate SSL methodology to BEATs as it does not use any tokenization. Numbers are reported on standard audio classification tasks like ESC-50 and AudioSet.

Efficient Audio Transformer (EAT): EAT [50] is another recently proposed SOTA audio encoder trained on AudioSet [23].

4.2. Evaluation Tasks

Table 3 lists all evaluation tasks. Linear probing measures the effectiveness of pretrained representations without any fine-tuning induced variations [53]. We benchmark linear probes on sound tasks using the X-ARES toolkit [2]. This toolkit converts all individual metrics to be in the range of 0-1, with higher values being better. The sound reasoning tasks evaluate the utility of audio representations even before integrating with frozen language models to build ALMs. We consider two setups: (1) a multi-modally trained text encoder, like CLAP [51], and (2) an independently trained text encoder, like BERT. By comparing these approaches, we can understand how an audio encoder will behave with audio-informed text embeddings. Therefore, our benchmark bridges the gap between low-level feature extraction and high-level cognitive reasoning, offering a structured framework to improve audio encoders for ALMs. For audio captioning, our setup consists of the frozen audio encoder and a trainable transformer decoder, initialized randomly [54].

[2] `https://github.com/jimbozhang/xares/tree/main`

Table 6: Bioacoustics: Full-finetuning evaluations on BEANS [40]. OpenBEATs achieves SOTA results performing competitively with models pre-trained with audio-text supervision [31], [51], in-domain SSL [14], [52], and SSL models with more parameters [25].

Model	Pre-training Data Instances	#Params	Sound Event Classification (acc)					Sound Event Detection (mAP)				
			Watkins	CBI	HumBugDB	Dogs	Bats	DCASE 21	Rfcx	Gibbons	Hiceas	Enabirds
Audio-text CL												
LAION-CLAP [51]	633k pairs	31M	89.1	62.2	82.0	96.4	75.3	47.0	14.5	29.6	62.7	66.4
BioLingual [31]	1.1M pairs	31M	89.4	74.4	81.7	97.1	76.6	47.5	17.8	37.6	67.7	68.8
Bioacoustics SSL												
AVES [52]	1.8M	89M	87.9	59.8	81.0	95.0	74.8	39.2	13.0	28.4	62.9	55.5
GPM-BT [14]	1.2M	89M	**91.4**	54.3	81.1	94.2	77.4	45.4	12.9	34.5	65.0	**62.4**
Dasheng [25]	97M	90M	77.3	63.1	69.9	95.7	75.5	45.6	11.7	41.8	43.8	43.9
		600M	79.7	67.8	73.6	95.7	77.3	45.5	13.6	47.3	58.8	45.6
		1.2B	79.1	66.7	72.2	**97.1**	76.5	46.4	13.2	36.8	57.4	45.0
BEATs												
iter 1	2M	90M	87.0	61.9	80.1	**97.1**	78.8	45.0	8.6	45.8	71.3	48.8
iter 2			87.9	64.4	87.3	92.1	79.3	45.2	10.6	44.0	**69.5**	51.1
iter 3			89.4	64.0	80.6	90.7	76.8	43.8	10.3	44.8	67.4	49.6
OpenBEATs Base												
iter 1	7.4M	90M	85.0	63.6	84.2	95.0	78.4	44.5	10.1	45.6	68.2	46.2
iter 2			86.4	64.9	87.3	96.4	77.3	46.2	13.2	47.3	68.2	49.0
iter 3			87.0	64.5	84.8	95.7	76.9	46.4	12.4	42.3	68.1	49.0
OpenBEATs Large												
iter 1	7.4M	300M	88.8	67.8	86.6	95.0	79.4	**49.5**	13.0	45.3	69.1	52.1
iter 2			88.8	68.8	**87.7**	95.7	**79.7**	47.6	13.0	**50.2**	68.3	53.4
iter 3			88.2	**69.4**	87.4	95.7	79.5	46.2	**14.9**	49.3	66.0	54.4

Table 7: Audio reasoning tasks. OpenBEATs outperforms other encoders on entailment and AQA and performs competitively with BEATs on captioning.

Model	AQA (BERT / CLAP)		Entailment		Clotho AAC
	Open	Yes-No	BERT	CLAP	
Metrics	acc	acc	acc	acc	CIDEr
BEATs	10.5 / 9.7	56.7 / 57.5	73.0	71.5	39.0
Dasheng-1.2B	10.4 / 10.1	58.5 / 56.7	**74.5**	73.0	**47.7**
OpenBEATs Base	12.2 / 11.1	59.4 / **59.4**	73.2	**75.2**	35.3
OpenBEATs Large	**12.7** / **11.5**	**59.9** / 58.7	73.2	74.2	37.5

Table 8: OpenBEATs improves performance on two of the three music tasks.

Model	GTZAN Genre	NSynth-I	NSynth-P
BEATs	87.1	79.9	**93.1**
Dasheng Base	87.1	80.9	90.9
OpenBEATs Base	88.1	79.5	92.5
OpenBEATs Large	**89.1**	**81.7**	92.7

5. RESULTS AND DISCUSSION

Linear probing on X-ARES: As summarized in Table 4, OpenBEATs `Large` achieves the best linear-probe accuracy on DESED and US8k, and remains within a few points of the 1.2 billion parameter Dasheng on FSD50K and ESC-50. Notably, it does so with four times fewer parameters than Dasheng-1.2B and a considerably smaller pre-training data budget. These results show that token prediction based modeling approach scales more efficiently, delivering competitive performance with less data.

Environmental Sound: OpenBEATs consistently delivers strong performance on the canonical ESC-50 and FSD-50K benchmarks in Table 5. The `Large` model attains 95.8 % accuracy on ESC-50 and 57.5 mAP on FSD-50K, confirming that parameter scaling pays off after multi-domain data saturates the 90 M scale. Performance gaps in AudioSet originate primarily from differences in the evaluation dataset, since YouTube is not a static source and our evaluation set differs by 5% from the one reported in BEATs. We verified the correctness of our training and inference code by replicating the numbers using the public BEATs inference codebase. We release both codebases and the YouTube IDs for our copy of the AudioSet publicly. The improved performance from OpenBEATs `Large` on AS-2M shows the benefits of multi-domain pre-training. These results demonstrate that the pre-training technique scales gracefully.

Bioacoustics: Section 4 shows that the OpenBEATs attains SOTA results on 6 of the 10 BEANS datasets among all SSL models. Even against billion-parameter Dasheng variants, OpenBEATs `Large` remains better confirming that performance scales favourably with masked token prediction based modeling applied to multi-domain datasets. Crucially, OpenBEATs also outperforms the text-supervised audio–text contrastive baselines like LAION-CLAP and BioLingual, demonstrating that scaled-up SSL can surpass models that have access to paired textual information. These findings show that multi-domain pre-training yields general-purpose representations.

Sound Reasoning and Music Tasks: OpenBEATs achieves best accuracies on both AQA and entailment tasks, underscoring the benefits of its larger capacity and broader pre-training corpus (Table 7). Consistently, we observe that scaling the model size and expanding pre-training data yields systematic gains for both Dasheng and OpenBEATs, confirming that representation quality improves with greater model and data scale. Nevertheless, OpenBEATs lags behind on audio-captioning. One possible reason is that we applied the recipe tuned for BEATs by [54] to other models without sufficient hyper-parameter exploration. OpenBEATs also improves performance on music tasks (Section 4), showing that multiple domains in pre-training data produce general purpose representations.

6. CONCLUSION

In this work, we open-source the complete pre-training pipeline for BEATs and demonstrate its effectiveness through strong performance across diverse benchmarks. Our comprehensive evaluation spans multiple domains and tasks, showcasing the robustness and versatility of the learned representations. We find that multi-domain pre-training enables the model to generalize effectively, achieving SOTA results on various datasets. These findings establish token-prediction based masked modeling on multi-domain datasets as a powerful framework for general audio representation learning.

7. ACKNOWLEDGEMENTS

This work used the Bridges2 system at PSC and Delta and DeltaAI system at NCSA through allocations CIS210014 and IRI120008P from the Advanced Cyberinfrastructure Coordination Ecosystem: Services & Support (ACCESS) program, supported by National Science Foundation grants #2138259,#:2138286, #:2138307, #:2137603, and #:2138296.

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